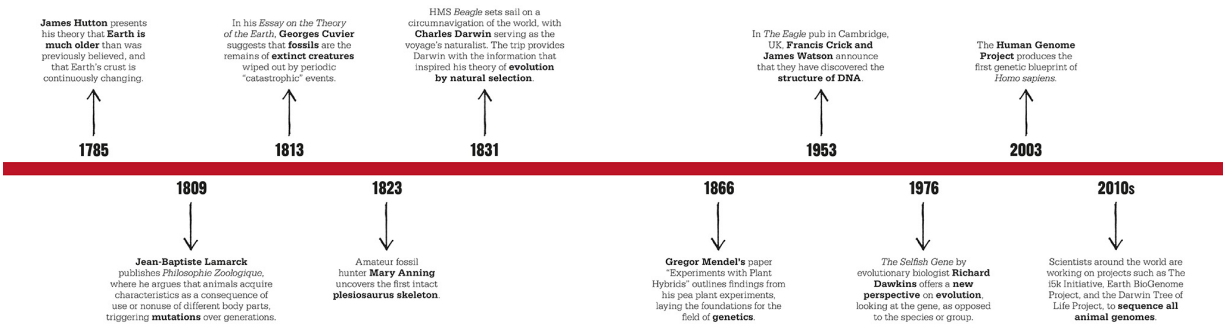


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Natural selection

In 1858, Charles Darwin and Alfred Russel Wallace delivered a paper that would change biology forever. Darwin's observations on the epic voyage of the *Beagle* (1831–36), his correspondence with other naturalists, and the influence of Thomas Malthus's writings inspired Darwin's insight that evolution came about by what he called natural selection. He spent 20 years gathering supporting data, but when Wallace wrote to him with the same idea, Darwin realized it was time to go public. His subsequent book, *On the Origin of Species by Means of Natural Selection*, provoked outrage.

Although the idea of evolution became widely accepted, the mechanism that made natural selection possible was not yet known. In 1866, an Austrian monk called Gregor Mendel made a huge contribution to genetics when he published his findings on heredity in pea plants. Mendel described how dominant and recessive traits pass from one generation to the next, by means of invisible "factors" that we now call genes.

The rediscovery of Mendel's work in 1900 initially sparked sharp debate between his supporters and many Darwinians. At the time, evolution was believed to be based on the selection of small, blending variations, but Mendel's variations clearly did not blend. Three decades later, geneticist Ronald Fisher and others argued that the two schools of thought were complementary, rather than contradictory. In 1942, Julian Huxley